Transport along a pair path in a doubly dependent family agrees with the composite of the two transports

The Univalent Foundations Program

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Editorial note. The counted statement is the author lem environment at source lines 1712-1723 and its proof environment opens on line 1724, the next line; the pairing is by adjacency and the retained proof range 1724-1726 is the whole environment.

Editorial note. The book label-number dump gives this label as 8.6.10 (env lem, chapter 8, section 6, tenth counter value), so the counters chapter 8, section 6 and thm 9 (zero-based) are set so that the item prints with the book number.

Editorial note. The statement cites one label outside this chapter, the transport of an arrow type, declared in external_references as a non-load-bearing navigation pointer: eq:transport-arrow in the basics chapter; the retained statement uses it only to name the first of its two transport steps.

Editorial note. The author proof is the two-line argument "By path induction, we may assume a2 is a1 and m is reflexivity, in which case both functions are the identity", quoted here verbatim; the card records that reduction as the whole proof of the comparison.

Editorial note. The retained preamble is the book own macro library and layout adapter as stored for this chapter; no package outside it is requested.

Editorial scope. One original result of the pinned HoTT Book source tree, counted once: the transport comparison used in the Freudenthal proof (author label thm:freudlemma, book Lemma 8.6.10).

Allowed standard prerequisites (editorial).

- Transport along a pair path. The transport of an element of a fibre of the uncurried family along a pair path is computed by a coherence path together with the transport in the second component.. It is the left-hand function compared in the statement.
- Univalence and the action of a dependent function on a path. Univalence turns an equivalence between types into an identification of them, and the action of a dependent function on a path computes the transport of its values.. They produce the equivalence on the right of the statement and justify the second step of the composite.

- Path induction. A statement about an arbitrary identification can be proved by proving the case of reflexivity.. It is the entire proof: after the reduction both compared functions are the identity.

Source reference. Original equation 2.9.4: sources/hott-basics/pinned-original-source.tex, source lines 1629–1629. Book equation 2.9.4, the transport rule for an arrow type, cited inside the retained statement to name the first transport step of the composite. Navigation pointer only.

Lemma 8.6.10. *Let $A : \mathcal{U}$, $B : A \to \mathcal{U}$, and $C : \prod_{(a:A)} B(a) \to \mathcal{U}$, and also $a_1, a_2 : A$ with $m : a_1 = a_2$ and $b : B(a_2)$. Then the function*

$$\mathsf{transport}^{\widehat{C}}(\mathsf{pair}^{=}(m,t), \text{–}) : C(a_1, \mathsf{transport}^{B}(m^{-1}, b)) \to C(a_2, b),$$

where $t : \mathsf{transport}^{B}(m, \mathsf{transport}^{B}(m^{-1}, b)) = b$ is the obvious coherence path and $\widehat{C} : (\sum_{(a:A)} B(a)) \to \mathcal{U}$ is the uncurried form of C, is equal to the equivalence obtained by univalence from the composite

$$\begin{aligned} C(a_1, \mathsf{transport}^{B}(m^{-1}, b)) &= \mathsf{transport}^{\lambda a.\, B(a)\to\mathcal{U}}(m, C(a_1))(b) && \text{(by (2.9.4))} \\ &= C(a_2, b). && \text{(by } \mathsf{happly}(\mathsf{apd}_C(m), b)) \end{aligned}$$

Proof. By path induction, we may assume a_2 is a_1 and m is refl_{a_1}, in which case both functions are the identity. □